Vision-Based Flow-Front Assessment in Vacuum-Assisted Resin Transfer Molding

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Our work presents four contributions towards the real-time tracking and control of VARTM processes. The first is an integrated classical computer-vision pipeline that combines a peak-brightness reference, a darken-only difference, region-of-interest restriction, dynamic-lag reference selection, morphological cleanup, persistence-based temporal locking, and run-time camera-shift registration, reaching mean mask Intersection-over-Union of 0.921 (95 % CI $[0.889, 0.943]$) and mean boundary F_1 of 0.433 (CI $[0.396, 0.473]$) on the benchmark introduced below. On the same benchmark, the two prior published classical-CV pipelines for VARTM flow-front segmentation, reimplemented from their source papers, reach mean boundary F_1 of 0.116 (Lekanidis and Vosniakos 2020) and 0.187 (Almazán-Lázaro 2022); their outputs lock onto distribution-medium pattern and bag-wrinkle artifacts rather than the resin front. The second is a component-removal ablation that characterizes the marginal IoU effect of each named primitive. The third is a regime-indexed configuration lookup that recommends preprocessing settings as a function of run circumstances so a practitioner can select an empirically-grounded starting point. The fourth is the benchmark itself, namely a 55-frame hand-labeled subset across 11 VARTM infusion runs with per-sample region-of-interest masks and a documented labeling protocol, against which the two prior pipelines above are evaluated as reference points and any future flow-front segmentation method can be evaluated identically. The pipeline runs at 30 frames per second on a single CPU and at 67 frames per second on a CUDA implementation for a 1920×1080 input.

I. Nomenclature

F_t	=	converted-colorspace frame at time index t
G_t	=	effective reference image at time index t
D_t	=	per-pixel response field at time t
R	=	binary region-of-interest mask
α	=	exponential-moving-average factor for running-reference mode
κ	=	sqrt-area growth factor estimated from calibration frames
ρ	=	target wet-fraction for dynamic reference mode
λ	=	user lag scale factor for dynamic reference
τ_t	=	elapsed time at frame index t in seconds
$\Delta\tau_t$	=	dynamic lag in seconds for the reference frame
IoU	=	Intersection-over-Union mask agreement score
F_1	=	boundary F_1 score, mean across pixel tolerances $\tau \in \{1, 3, 5\}$

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II. Introduction

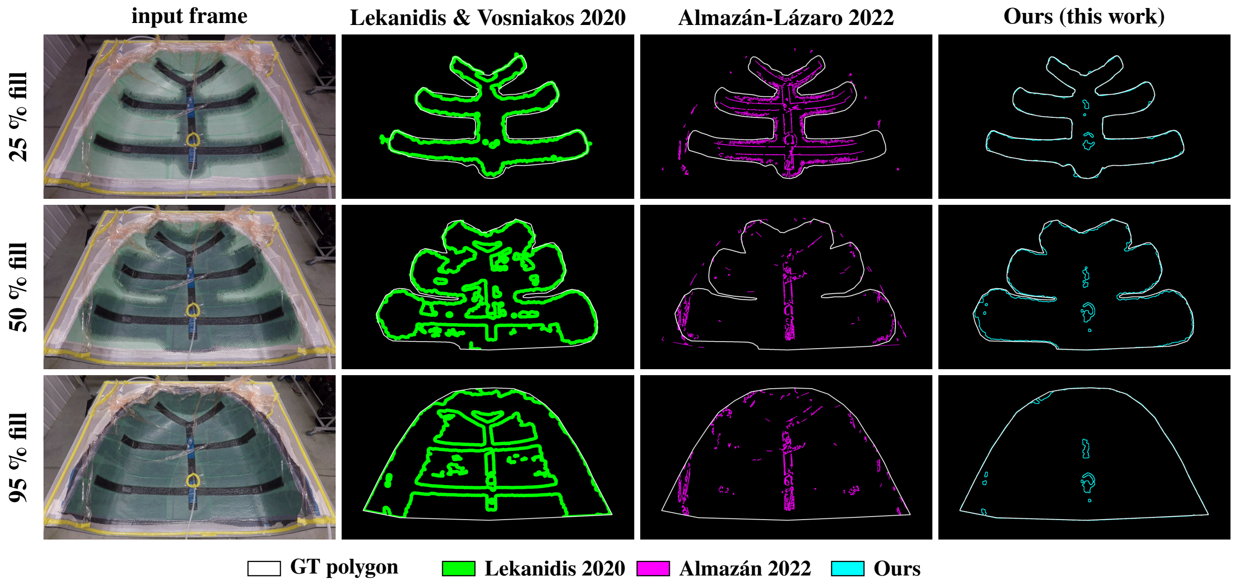


Fig. 1 Per-frame boundary comparison on the canonical reference video at three fill positions (25%, 50%, 95%). Region IoU and boundary F_1 for the same comparison are reported in Table 10.

VARTM, or Vacuum Assisted Resin Transfer Molding, is a widely used process for one-off composite laminates in research, marine, wind, and large-format defense applications [1,2], and is the principal route for parts whose size or geometry rules out autoclave processing [3–5]. However, the process is susceptible to defects that can render parts unusable. Dry spots from incomplete impregnation [6], race-tracking along edges and the distribution medium [7], voids and porosity from trapped air [8], and resin starvation in interior regions of the laminate [9] can all reduce stiffness and strength of the cured part [10], which is often undesirable in production. Identifying, predicting, and preventing these defects spans post-cure non-destructive evaluation, in-situ embedded sensing during infusion, forward Darcy-flow simulation of the run, and digital-twin systems that combine these signals. Each modality contributes a different slice of evidence, but most are ultimately reasoning about which fabric wetted, where, and when. The resin flow front therefore consolidates much of this information into a single quantity that can be measured directly during the run. Real-time, precise information about the front position gives the operator the most direct lever on part quality during an active infusion [11,12]. Given that signal, the operator can re-sequence inlets to redirect flow toward a starved region [13,14], open additional vents to break a race-track [15–17], or stop the infusion before resin gels [18].

Deriving or measuring the position, shape, and rate of the flow front is non-trivial, since the front is a two-dimensional moving boundary inside an opaque lay-up, and any sensing modality must trade spatial resolution against intrusiveness in the part. Distributed dielectric arrays [19] and area-sensor grids [20] produce excellent local wetness measurements at every electrode or cell, but both options require sensors embedded in the lay-up and so alter the local permeability of the part they measure. Fiber-optic frequency-domain reflectometry [21] time-stamps resin arrival at every grating along an embedded fiber resulting in excellent temporal precision, but the interrogator instrumentation is expensive enough to be impractical for one-off research panels, and front shapes between gratings must be interpolated. Vision-based methods, like a camera looking down through a transparent vacuum bag, provide dense pixel-level information about the flow front, however it only sees the top ply visible through the bag, so a dry spot one ply deep can go undetected. Additionally, the visible-light contrast between wetted and dry fabric depends on the specific resin-fabric pair, the bag transparency, and the lighting at the bench, making contrast non-stationary across runs [22]. Thermal-imaging variants [5] partially recover through-thickness saturation by reading the exotherm of curing resin, but the entry cost of a research-grade infrared camera is one to two orders of magnitude above the visible-light option, and the calibration is sensitive to ambient temperature drift.

Vision-based systems are the focus of this paper because a camera does not affect the flow of the part it observes, applies to any geometry without re-instrumentation, and in most labs is already being used as a procedural artifact at near-zero marginal cost. In vision-based systems, like thermal or RGB, the primary goal remains to retrieve either the current edge of the flow front or the entire flow front mask. The dominant approach to derive the front from the live video feed involves computing the change in pixel intensity as the resin wets the fabric, thresholding the response into a binary candidate mask, and cleaning the mask with morphology [23–25]. However, despite that deceptively simple description, the compute pipeline for this change in pixel intensity is very sensitive to disturbances and prior camera-CV pipelines disagree on the details that control how robust the model is to disturbances. Some systems clip the difference so that only darkening counts as wetting and others do not [25]. Some track a per-pixel running maximum as the reference and others pin a single early frame [23]. Some impose a temporal-locking rule that holds a positive detection through transient dropouts and others operate frame-by-frame [26].

Recent vision-based work has begun to replace these classical operators with learned detectors [26–30] and a parallel thread applies deep models to adjacent VARTM observables such as cure temperature distribution [31]. However, ML in this domain stalls on data scarcity since each VARTM run is its own regime of fabric, bag, lighting, and geometry [32]. Complicating matters further, no shared labeled video benchmark spans multiple molds, so a detector trained on one bench cannot be measured on the next [28]. The classical-CV pipelines deploy without per-bench training data, but they too have not been evaluated against any shared video benchmark, leaving open how they compare to each other or to any future method.

This paper closes these gaps with an integrated classical-computer-vision pipeline evaluated on a 55-frame hand-labeled benchmark spanning eleven distinct VARTM infusion runs. Its four contributions are the following.

1. An integrated VARTM flow-front segmentation pipeline that combines peak-brightness reference, darken-only difference, ROI restriction, dynamic-lag reference selection, Otsu-plus-offset thresholding, morphological cleanup, persistence-based temporal locking, and run-time camera-shift registration in one configuration, with mean Intersection-over-Union (IoU) 0.921 (95% bootstrap CI $[0.889, 0.943]$) and mean boundary F_1 0.433 (CI $[0.396, 0.473]$) on a 55-frame hand-labeled subset spanning eleven distinct infusion runs.
2. A regime-indexed configuration lookup, derived from the per-sample ablation below, that recommends preprocessing settings for each pipeline component as a function of run circumstances. The lookup is the practitioner-facing translation of the per-sample evidence and is what an operator on a different bench would consult before tuning their own infusion.
3. A per-sample component-removal ablation on the same 55-frame subset that quantifies the marginal IoU effect of each named primitive, both as an eleven-sample mean and as a per-sample breakdown. The breakdown is reported in full so the joint composition is justified by the per-sample evidence rather than by a single mean. The ablation is the empirical content that the pipeline and the lookup both rest on.
4. A 55-frame hand-labeled benchmark spanning eleven distinct VARTM infusion runs, with per-sample region-of-interest masks and a documented labeling protocol. The two prior published classical-CV pipelines for VARTM flow-front segmentation, Lekanidis and Vosniakos 2020 [23] and Almazán-Lázaro 2022 [25], are reimplemented from their source papers and evaluated on this benchmark as reference points.

III. Related Work

This work builds upon four research pillars of the composite-manufacturing and computer-vision literatures. The first pillar is the forward and inverse physical modeling of resin flow in Liquid Composite Molding (LCM) and Vacuum-Assisted Resin Transfer Molding (VARTM), which predicts the per-pixel front from a permeability field or recovers a permeability field from observed front geometry. The second pillar is the embedded-sensor observation of the front during infusion, which produces dense in-plane wetness signals at an instrumentation cost that scales with sensor density. The third pillar is the real-time control of resin infusion, where the front signal drives valves, vents, or pumps, and where a recent camera-based subset replaces point sensors with computer vision. The fourth pillar is the classical computer-vision tradition of thresholding, morphology, image alignment, and topological border-following, whose named operators the integrated pipeline composes into a per-frame per-pixel mask producer. The remainder of this section addresses each pillar in turn. The third pillar's camera-based subset is the closest prior art and receives its own subsection while the final subsection attributes the operators of the fourth pillar to their canonical sources and names the specific evidence gap this paper supplies.

A. Modeling, sensing, and control of the flow front

The VARTM flow-modeling literature traditionally treats the flow front as a given input. Forward simulators predict where resin will flow across the laminate given material permeability and inlet conditions, and have been validated against experiments on flat and stiffened plates [33,34]. These geometries are simpler than the multi-section preforms used in production parts, which limits how directly the simulator results transfer to operational benches. Conversely, inverse methods work backward from the observed front. One approach combines overhead camera measurements of how fabric thickness varies across the laminate with a center-injection experiment to recover a 2D permeability map [35]. Another method compares live pressure-sensor data against a closed-form Darcy-flow model of the moving front to adjust injection parameters on the fly [36]. Learning-based variants extend the same thread, using piezoelectric sensors and a hybrid learned model to identify and forecast the front [29], or physics-informed neural networks to recover permeability fields from simulated Darcy data [37]. A related line of work trains surrogate models to predict the 2D front map from sparse pressure sensors on simulated runs [27], later extending the same approach to material-property inference under simulation-to-real transfer [28]. Every method discussed above depends on a front map, which is currently obtained from finite-element simulation or manual annotation of plate experiments. However, simulation and hand-labeling are slow and bench-specific, which prevents real-time front estimation during an active infusion. The integrated pipeline we introduce produces that map directly from production video, removing both bottlenecks.

The observation literature measures the front directly but does so through instrumentation that discretizes it, traditionally through modalities like dielectric, thermal, ultrasonic, fiber-optic, and electrical-resistance [5]. Dielectric tracking gives unsaturated and saturated front positions at each electrode pair at an installation cost that is hard to beat, but the spatial resolution of the resulting front is bounded by the electrode pitch [19]. Area-sensor arrays dramatically improve coverage by tiling a polyimide film into a dense capacitive matrix and recovering the impregnated-area ratio per cell, which is the closest sensor-side analogue to a full-field flow-front mask [20]. However, the resulting front is still a quantized grid limited by the cell size, and the polyimide film must also be embedded in the lay-up, which is intrusive to install in production tooling and alters the local permeability of the part being measured. Fiber-optic strategies use Fiber Bragg Grating arrays or Optical Frequency Domain Reflectometry to time-stamp resin arrival at every grating along an embedded fiber, with recent distributed-sensing implementations pushing toward life-cycle structural-health monitoring of large Carbon-Fiber-Reinforced-Polymer (CFRP) parts [21]. These methods deliver excellent temporal precision and can survive infusion and cure, but the same tradeoff between sensor density and front accuracy reappears, since arbitrary front shapes between gratings are interpolated rather than measured. Distributed carbon-nanotube textile sensors push the spatial-coverage axis further by integrating the sensing layer into the lay-up itself [38], but the textile becomes a permanent part of the cured laminate and adds its own material to the stack-up. Across all of the aforementioned methods, the resolution of the resulting front is limited by the spacing of the sensing elements, and the approaches that come closest to dense coverage do so by embedding the sensor into the part being measured. The integrated pipeline we introduce avoids both costs by reading the front with a single overhead camera.

The control literature is often the largest downstream user of the flow front data, since it uses the front signal to act on the process in real time. Early closed-loop systems coupled point-sensor arrival times to a finite-element flow model so that multi-segment injection valves could be re-sequenced to chase the front toward starved regions and away from race-tracks [39]. Subsequent sensor-based controllers refined the same architecture to handle race-tracking, dry-spot prevention, and permeability variation through paired sensing and actuation [11,12,14–17]. A more recent camera-based subset replaces point sensors with a top-mounted camera, segments the front from the live video stream with a classical pipeline of difference-and-morphology operators, and toggles inlet valves on the resulting front position [23–25]. The camera-based subset is the most promising of these for closed-loop control, because a single overhead camera gives the controller a per-pixel front signal at every frame without perturbing the flow it observes or degrading the structural performance of the final part. Recent fusion work supports this trend, using a learned You-Only-Look-Once (YOLO) detector on camera video as the primary segmenter and adding dielectric channels only as redundancy [26].

Adjacent to these three pillars is a post-cure non-destructive-evaluation literature and a closed-loop digital-twin literature, both of which the integrated pipeline does not build on directly but which sit naturally downstream of it. Defect characterization in cured composite parts reads the trace of the front after the fact. Ultrasonic C-scan recovers delamination, voids, and porosity at sub-millimeter resolution [40], infrared thermography covers larger areas at lower spatial resolution and without contact, and fusion of C-scan with thermographic phase imaging measurably improves shallow-defect signal-to-noise on CFRP laminates [41]. The defects these methods detect are the consequence of a particular flow trajectory, and the per-frame mask produced by the integrated pipeline describes that trajectory directly while it is still acting. A complementary digital-twin thread fuses in-situ sensor data, forward simulation, and learned controllers to identify defect signatures in real time and to use model-predictive control over vacuum and thermal cycles to suppress dry-spot formation [42]. Such systems are candidate consumers of the per-pixel front signal at the sensing layer, since the wet-versus-dry observable they require is exactly what the pipeline emits.

B. Camera-based front segmentation

The closest prior art to the integrated pipeline is the camera-CV subset of the control family above, and the contribution of this paper is best located against it. Lekanidis and Vosniakos built a webcam-plus-microcontroller rig for Vacuum-Assisted Resin Infusion (VARI) that segments the resin flow front with a Matlab pipeline of Gaussian blur, grayscale conversion, contrast stretch, Otsu binarization, closing on a ten-pixel disk, Sobel edge detection, opening with a one-hundred-twenty-pixel area threshold, and dilation, and toggles inlet valves on the resulting front position [23]. Almazán-Lázaro closes a similar loop on resin infusion, regulating flow-front velocity to an experimentally-determined optimum with a Proportional-Integral-Derivative (PID) controller over a pipeline that combines region-of-interest cropping, Scaramuzza-style lens-distortion correction, histogram equalization, first-frame absolute differencing, grayscale conversion, a five-by-five mean filter, Sobel-gradient segmentation, paired erosion and dilation, and small-area removal [24,25]. Esposito and colleagues replace the classical segmenter with a learned YOLO detector trained on labeled video, fuse the output with dielectric channels for redundancy, and demonstrate real-time front monitoring and prediction on a single-fabric study [26]. The three systems demonstrate that vision alone, or vision plus dielectric, can drive VARTM monitoring and control on the bench at which they were instrumented.

What the three systems do not report is the question this paper answers. They report end-to-end behavior, never the joint or individual contribution of the segmentation primitives in use. They treat the per-frame mask producer as a unit that succeeds or fails as a whole rather than as a composition of named operators whose marginal contributions can be measured. They disagree on details that propagate directly into how robust each pipeline is to disturbance. The Almazán-Lázaro pipeline clips the difference so that only darkening counts as wetting and the Lekanidis-Vosniakos pipeline does not. The Lekanidis-Vosniakos pipeline tracks a per-pixel running maximum as the reference and the Almazán-Lázaro pipeline pins a single early frame as a first-frame absolute differencing reference. The Esposito pipeline imposes a learned temporal regularization through the YOLO detector trained on sequential frames, and the two classical pipelines operate frame-by-frame on the segmentation side. None of these disagreements has been adjudicated on a labeled multi-mold benchmark because no such benchmark exists in the published literature. Each prior paper reports excellent agreement on the single mold it was developed for, and the reader is left to guess which knob settings would carry over to a different bench.

Adjacent to the front-segmentation literature is a broader composite-defect-detection literature that increasingly leans on deep convolutional and transformer detectors trained on hand-curated defect crops. A recent material-defect-detection survey identifies pixel-accurate labels as the primary bottleneck, ahead of model capacity [43]. Modern detectors such as YOLO [44,45] and segmentation architectures such as U-Net [46] illustrate why this matters, because they are starved for in-domain pixel labels rather than for parameters. A per-pixel front-mask producer that runs deterministically on production video, as opposed to a learned detector that requires its own labeled-video training set per bench, is therefore upstream of that bottleneck. The integrated pipeline of this paper is positioned to fill that upstream role, and the 55-frame hand-labeled benchmark introduced alongside it is positioned to support evaluation of any future learned segmenter that wants to claim improvement over the classical baseline.

Two specific operators define the difference between the prior camera-CV pipelines and the integrated pipeline. The lens-distortion calibration step applied once at setup time in Almazán-Lázaro [47] is the only operation in either prior camera-CV pipeline that has no analogue in this work, and the run-time camera-shift registration described in Section 3 is the only operation in this work that has no analogue in either prior camera-CV pipeline. The two are complementary axes rather than substitutes, and a deployed system would benefit from both.

C. Classical computer-vision primitives

The integrated pipeline stays deliberately within well-understood classical operators because the empirical claim of this paper is per-component, and a learned segmenter would conflate the contributions of the components inside it. Every named stage maps to a primitive with a canonical attribution and a documented behavior, and the per-component ablation reported in Section 5 is well defined precisely because each stage can be disabled independently. Wet-versus-dry separation in each frame is performed by Otsu's between-class-variance threshold [48], with the Triangle method [49] exposed as an alternative when one class dominates the histogram. Camera-shift registration combines phase correlation [50] on a Hanning-windowed downsample with iterative Enhanced Correlation Coefficient (ECC) refinement [51], run on a static-edge mask of the current region of interest so the warp is driven by mold and frame edges that do not move with the wet front. Polygon export to Common Objects in Context (COCO) and YOLO formats uses the Suzuki-Abe topological-border-following algorithm [52], optionally simplified by Douglas-Peucker reduction at a user-supplied tolerance [53]. The peak-brightness reference, the darken-only clip, the post-delta blur, the paired close-and-open morphology with an area-floor connected-components filter, and the count-based temporal lock are per-pixel compositions of arithmetic and comparison whose value lies in the joint configuration in which they appear rather than in any single canonical attribution.

None of these primitives is novel in isolation. The contribution of this paper is the joint composition, the per-stage parameterization, and the per-component evidence that the joint matters on a labeled multi-mold subset. The four pillars on which this work builds each touch the flow front from a distinct angle, and none on its own supplies the evidence this paper provides. The modeling pillar consumes a front map it does not measure, and obtains the map from simulation. The observation pillar measures a front it cannot densely cover, and bounds front fidelity by sensor pitch. The control pillar acts on a front it produces with a segmenter it does not characterize per component. The classical computer-vision pillar supplies the named primitives but does not on its own evaluate any composition of them on a multi-mold VARTM benchmark. The camera-CV subsystems closest to this work, Lekanidis-Vosniakos and Almazán-Lázaro, differ in which subset of the classical primitives they use and report only end-to-end control behavior on a single mold. None reports the joint contribution of the components in use, none reports the marginal cost of any component left out, and none evaluates against a labeled video benchmark that spans more than one mold. This paper supplies that evidence on a 55-frame hand-labeled subset spanning eleven distinct infusion runs, with the component-removal ablation reported both as an eleven-sample mean and as a per-sample breakdown.

IV. Method

A. Pipeline overview

Unlike contemporary ML models, the detection algorithm here treats each video frame as a comparison against a reference image of the dry preform. Where resin has wetted the fabric, the local appearance darkens, and the absolute change is largest near the advancing flow front. The pipeline first checks for camera motion against the previous frame and, when motion crosses a configured threshold, registers the live frame back into the reference coordinate system before any difference is computed. The registered frame is optionally smoothed with a pre-delta blur whose output feeds both the running peak map and the difference computation, so the reference and the comparison see the same bandwidth-limited input. The pipeline computes the per-pixel difference inside a rectangular region of interest, clips it to admit only darkening so that specular highlights from the vacuum bag are rejected, normalizes the resulting field, thresholds it into a binary candidate mask, cleans the mask with morphological closing and opening, removes small connected components, and finally locks pixels whose wet label has persisted for several consecutive frames. The locked mask is the canonical output, the optional heatmap renders the underlying delta field for visual diagnosis, and the optional overlay draws the mask boundary onto the original Vacuum-Assisted Resin Transfer Molding (VARTM) frame for human review. Figure Fig. 2 shows the sixteen stages in order.

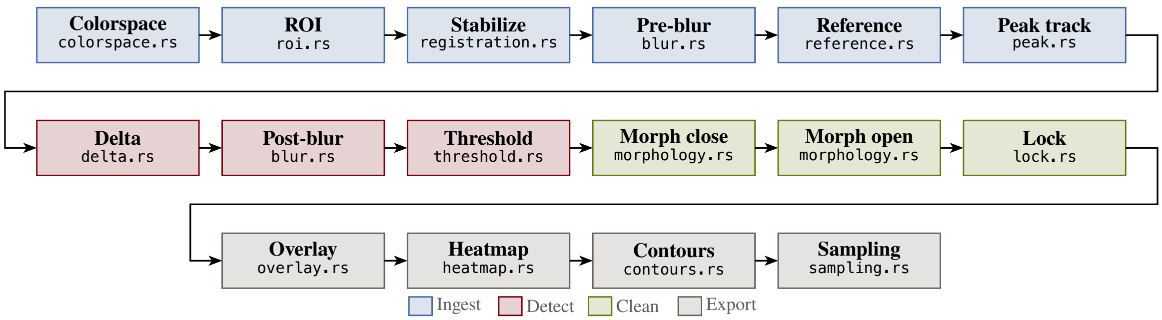


Fig. 2 Sixteen-stage integrated pipeline. Each frame flows from decode through an optional camera-shift registration and pre-delta blur, then through an appearance-difference comparison against a chosen reference, a post-delta blur, a threshold, paired morphological closing and opening passes, and finally a temporal lock that stabilizes the boundary across frames. The same mask drives the binary, heatmap, and overlay renderers.

The pipeline is deliberately classical because the empirical claim of this paper is per-component. Every stage is an explicit function of one frame, the chosen reference image, and a small set of scalar parameters, which is the property that makes the component-removal ablation in Section 5 well defined. A learned segmenter would conflate the contributions of the components inside it, and the question this paper asks would not be answerable.

B. Frame decode and colorspace conversion

The first stage decodes each Blue-Green-Red (BGR) frame from the input video. The second projects that frame into a working colorspace. The pipeline supports four options, namely the International Commission on Illumination (CIE) 1976 $L^*a^*b^*$ colorspace (CIELAB), Red-Green-Blue (RGB), Hue-Saturation-Value (HSV), and 8-bit grayscale. The integrated configuration uses CIELAB because for the resin-and-fabric combinations in the eleven samples evaluated here, wetting primarily darkens the lightness channel L^* rather than shifting chrominance, so the single-channel L^* projection used in the difference computation captures most of the signal. The choice is regime-dependent rather than universal. A pigmented resin or a colored fabric could shift the balance of evidence into chrominance and make RGB or HSV preferable, which is why the alternatives remain selectable. The eleven-sample mean and per-sample breakdown of the colorspace effect are reported in Table Table 13. We denote the converted frame at index t by F_t.

Colorspace projection placeholder. Four-panel grid showing the same canonical input frame projected into each of the four supported colorspaces. Column 1: raw BGR. Column 2: CIELAB lightness L^* (used by the integrated configuration). Column 3: HSV value channel. Column 4: 8-bit grayscale. Each panel rendered with a perceptually uniform colormap so wet-vs-dry separability per projection is visible. A bottom strip shows the per-projection histogram for the laminate region of interest with the wet and dry modes annotated. Regenerated from the new ablation runs.

Fig. 3 The four working-colorspace projections supported by the pipeline.

Table 1 Working-colorspace options selectable via `--colorspace`.

Option	Description
CIELAB	CIE 1976 $L^*a^*b^*$. Wetting primarily darkens L^* for the resin-and-fabric combinations evaluated in this paper.
RGB	Red-Green-Blue. Useful when chrominance shifts under wetting are visible per channel, e.g. pigmented resin.
HSV	Hue-Saturation-Value. Saturation tracks wet-out separately from luminance, useful under variable lighting.
Grayscale	8-bit luminance projection. Cheapest but loses chrominance evidence entirely.

C. Region of interest

The region of interest is a binary mask R that is one inside the laminate and zero elsewhere; all subsequent stages operate only on pixels where $R = 1$. The pipeline supports three forms for constructing R, mutually exclusive, selected by which configuration parameter is supplied. The rectangular form takes four fractional margins, one per edge, each clamped to the closed interval from zero to forty-nine hundredths of the corresponding side, and builds the rectangle bounded by those margins. A single configuration parameter sets all four margins symmetrically, with per-edge overrides available. The rectangular form is appropriate for laminates that fit a rectangular bounding box with no internal fixtures, which describes every sample in the labeling subset of Section 4. The imported-PNG form reads a single-channel grayscale image at the source video's resolution from a path supplied via `--roi-mask`, thresholds it at 127, and uses the resulting binary image directly as R. The imported-PNG form is appropriate when the laminate is non-rectangular (for instance a curved part boundary) or when fixtures, sensors, or labels inside the bag must be excluded from the difference computation but lie inside any axis-aligned rectangle that contains the laminate. The imported-COCO form reads a Common Objects in Context (COCO) JSON file from the same `--roi-mask` flag, locates the image entry whose `file_name` matches the input video, and rasterizes every polygon annotation on that image into R. The imported-COCO form is appropriate when the laminate boundary is already available as a polygon in an existing labeling project. The integrated configuration uses the rectangular form with `roi-margin = 0.15`; the imported forms are exercised in the per-mold tuning regimes described in Section 7.

ROI placeholder. Two-panel comparison of the two ROI forms on the same frame. Left: rectangular form. Single canonical input frame with the rectangular ROI mask R drawn as a translucent garnet overlay; the manifold flange and bag wrinkles outside the rectangle are visibly excluded, and a small inset shows the four fractional margins used in the integrated configuration. Right: imported-mask form. Same frame with a non-rectangular laminate boundary drawn as a translucent garnet overlay, with an internal fixture (bagging-tape patch, embedded thermocouple, or operator hand) drawn as a hatched exclusion region inside what would have been the rectangular bounding box. The imported form covers regimes the rectangular form cannot.

Fig. 4 The two equivalent forms for constructing the ROI mask R: a rectangle parameterised by four fractional margins (left), or an externally-supplied binary image at the source resolution (right).

Table 2 The two forms of the region-of-interest mask R. The integrated configuration uses the rectangular form.

Option	Description
rectangular	Bounding rectangle parameterised by four fractional margins (`roi-margin`, optionally overridden per edge by `roi-margin-top`, `-bottom`, `-left`, `-right`), each clamped to $[0, 0.49]$.
imported PNG	Single-channel grayscale image read from `--roi-mask` at the source video's resolution, thresholded at 127 to produce R. Used when the laminate is non-rectangular or when interior fixtures must be excluded.
imported COCO	Polygon annotations read from a `--roi-mask` JSON file in COCO format, image entry matched by `file_name` against the input video, polygons rasterized into R. Used when the laminate boundary is already labeled as polygons in an existing project.

D. Camera-shift detection and registration

A bumped tripod, a thermal expansion of the rig, or a hand brushing the camera in mid-run shifts the projected position of every laminate pixel by some vector that has nothing to do with wetting. The reference frame stops aligning with the live frame, the difference field lights up along high-contrast laminate edges, and the threshold catches that as wet. The pipeline detects the shift and corrects it before the difference is computed. For each frame, a single phase-correlation step on a fixed-resolution downsample of the working channel of the previous and current frames returns a translation $\left(d_x, d_y\right)$ and a confidence score. When either the per-frame magnitude $\sqrt{d_x^2 + d_y^2}$ or the cumulative drift across a five-frame rolling window exceeds the configured threshold, an iterative-coplanar-correlation refinement fits a translation or affine warp W_t on a static-edge mask of the current ROI, and the live frame is warped through W_t into the reference coordinate system before all downstream stages. The static-edge mask is recomputed at each shift event so the registration is driven by mold and frame edges that do not move with the wet front rather than by the wet region itself. The peak map is either discarded (`peak-on-shift = reset`, used by the integrated configuration), so the post-warp pixels do not accumulate against pre-warp brightness, or warped through W_t into the new coordinate system (`peak-on-shift = warp`), so the running maximum is preserved at the cost of carrying any residual registration error into the peak.

Camera-shift event placeholder. Three panels of the bumped-tripod event near frame 71 of the canonical reference video. Left: pre-event frame (frame 70). Center: post-event frame (frame 72) overlaid translucently on the pre-event frame with the phase-correlation translation vector drawn as an arrow. Right: the post-event frame after the iterative-coplanar- correlation refinement warps it through W_t back into reference coordinates, with the static-edge mask overlaid showing which mold-and-frame edges drive the warp. A sub-panel below plots the rolling five-frame cumulative motion magnitude across the run with a marker on frame 71. Regenerated from the new ablation runs.

Fig. 5 A bumped-tripod event in the canonical reference video. The phase-correlation step detects the shift, the iterative-coplanar-correlation refinement warps the live frame back into reference coordinates, and the rest of the recording shows only sub-pixel residual motion.

Table 3 Camera-shift handling options. The integrated configuration uses `motion-model = affine` and `peak-on-shift = reset`.

Parameter	Option	Description
`motion-model`	translation	Two-degree-of-freedom translation warp; cheaper to fit, recovers pure tripod nudges.
`motion-model`	affine	Six-degree-of-freedom affine warp; tolerates small rotations and shears in addition to translation.
`peak-on-shift`	reset	Discard the peak map at registration and restart accumulation from the warped frame.
`peak-on-shift`	warp	Warp the peak map itself through W_t into the new coordinate system; preserves the running maximum at the cost of carrying residual registration error.

E. Peak-brightness reference

The pipeline accumulates a running maximum of the working channel across all frames seen so far. When the pre-delta blur is enabled the maximum is updated from the blurred working channel rather than the raw channel so the peak map and the delta input share the same bandwidth, which prevents a one-pixel halo of speckle from shifting the peak above the eventual delta input and creating a phantom darkening signal. The motivation for the running maximum is that, on infusions where the dry preform is the brightest state the fabric reaches in the working channel, a per-pixel maximum tracks lighting drift caused by lamp warm-up and bag deformation rather than treating that drift as wetting evidence. Figure Fig. 6 illustrates the regime on one tracked pixel of the canonical input clip. The raw L^* value drifts upward as the lamps stabilize, the running peak P tracks that drift, and the front arrival separates as a sharp drop of more than thirty units below the peak. A fixed reference at frame zero would have included the post-warm-up lift in its difference. The assumption that wetting is the only event that lowers the peak does not hold universally. A specular flash that drives the peak above its long-term value, or a fabric whose wet state is brighter than its dry state for some channel, can produce a peak map that misrepresents subsequent wetting evidence. The IoU effect of disabling the primitive on the eleven-sample subset is reported in Table Table 13.

Peak-reference trace placeholder. One pixel of the canonical reference video tracked across all frames; raw L^* in garnet, running peak P in atlantic, with the front-arrival drop annotated. Regenerated from the new ablation runs.

Fig. 6 One pixel of the first canonical input clip tracked across all 706 frames. The raw CIELAB lightness L^* drifts up by roughly twenty units before the front arrives, and the running peak P absorbs that drift. When the resin reaches the pixel, L^* collapses by more than thirty units below the peak, which is the signal the difference stage detects. A single fixed reference at frame zero would have treated the warm-up brightening as a (negative) wetting event.

The peak map can be disabled through a single configuration parameter. The "no peak-brightness reference" row of Table Table 13 reports the IoU effect of doing so on each of the eleven samples, with the per-sample breakdown showing that the effect is not constant across infusion regimes.

F. Reference selection

The reference image G_t used for the difference computation has five modes. The first-frame mode pins $G_t = F_0$ for every t. The running mode updates an exponential moving average $G_t = (1-\alpha)G_{t-1} + \alpha F_t$ with $\alpha = 0.05$. The previous-frame mode uses a fixed-offset history $G_t = F_{t-k}$, with the buffer bootstrapped from F_0 until $t > k$. The absolute mode pins G_t to a user-specified absolute frame index. The dynamic mode adapts the lag online from a square-root-area growth model fit to the early frames. The integrated configuration uses the first-frame mode, which, combined with the peak map above, gives a piecewise-static reference whose only adaptation is the peak update.

The dynamic mode warrants a closer look because the integrated configuration does not use it for the headline numbers and disabling it is one of the rows in Table Table 13. For the first ten frames after detection bootstrapping (the calibration window), the pipeline records the wet area a_t inside the region of interest and the elapsed time τ_t at frame t. It then estimates a growth-rate factor κ as the median of $\frac{a_t}{\tau_t^{\frac{3}{2}}}$ across the calibration frames. The three-halves exponent is motivated by the area-growth law for radial Darcy infusion, where wetted area scales approximately as time to the three-halves power once flow is well established; the assumption holds approximately for the early-fill regime of the eleven samples in the labeling subset and breaks for race-tracking-dominated runs, which is why the dynamic-calibration window is sensitive to anomalous early frames as discussed in Section 7. Given κ and a target wet fraction ρ of the region-of-interest area $|R|$ (held at 0.2 in the integrated configuration), the dynamic lag $\Delta\tau_t$ in seconds that places the reference at the moment when the wet area was a fraction ρ of the current area is

$$\Delta\tau_t = \lambda \cdot \left[\left(\frac{\rho\ |R|}{\kappa} + \sqrt{\tau_t} \right)^2 - \tau_t \right], \tag{1}$$

clipped to be non-negative and scaled by a user lag factor λ. The reference frame is then read from a small cache at the integer index whose elapsed time is closest to $\tau_t - \Delta\tau_t$, falling back to the first frame whenever the calibration has not yet produced a finite κ. A linear-mode override replaces the sqrt-area growth fit with a linear lag schedule parameterised by `dynamic-lag-linear-start` and `dynamic-lag-linear-max`, which steps the reference frame back at a constant rate independent of the calibration estimate. The override is intended for diagnostic comparisons where the sqrt-area assumption is suspect, and a per-frame log of the chosen lag is written to `dynamic-lag-log` for post-hoc inspection.

Reference-mode comparison placeholder. Six-panel grid showing the same canonical input frame at fill position 50% under the five reference modes plus the dynamic linear-lag override. Top row: first-frame reference, running EMA reference, previous-frame fixed-offset reference. Bottom row: absolute-frame reference, dynamic sqrt-area reference, dynamic linear-lag override. Each panel shows the resulting D_t scalar field in the Turbo colormap so the reader can see how each reference mode selects different evidence on the same frame. The integrated- configuration mode (first-frame combined with the peak map) is bordered in garnet. Regenerated from the new ablation runs.

Fig. 7 The five reference-selection modes the pipeline supports plus the linear-lag dynamic override. The integrated configuration uses the first-frame mode combined with the peak map.

Table 4 Reference-selection options selectable via `--ref-mode`.

Option	Description
first	Pin $G_t = F_0$ for every t. Combined with the peak map, used by the integrated configuration.
running	Exponential moving average $G_t = (1-\alpha)G_{t-1} + \alpha F_t$.
previous	Fixed-offset history $G_t = F_{t-k}$.
absolute	Pin G_t to a user-specified absolute frame index.
dynamic (sqrt-area)	Adapt the lag online from a square-root-area Darcy growth fit on a calibration window of frames per Eq. Eq. (1).
dynamic (linear)	Constant-rate lag schedule, independent of the growth fit; intended for diagnostic comparisons.

G. Delta computation

Stage seven produces the per-pixel scalar field D_t that drives all downstream decisions. The integrated configuration's darken-only mode operates on the working (first) channel and records how much darker the frame is than its reference,

$$D_t(y, x) = R(y, x) \cdot \max(0, w_0 \cdot (G_t(y, x) - F_t(y, x, 0))), \tag{2}$$

where the reference G_t is the running peak map (per-pixel maximum of the working channel across all prior frames) when peak-reference is enabled, and the working-channel slice of the frame chosen by the reference-selection mode otherwise. The clip to non-negative values discards every pixel that becomes brighter than the reference, which removes specular flashes from the silicone vacuum bag and from condensation, neither of which are wetting events. A full-color mode replaces the difference with the channel-weighted Euclidean distance across all channels and is intended for HSV and RGB workflows where chrominance shifts are diagnostic. The per-channel weights w_0, w_1, w_2 are exposed via `channel-weights` and equal $1, 1, 1$ in the integrated configuration; non-uniform weights are appropriate when one channel of the working colorspace carries the wetting signal more strongly than the others. Figure Fig. 8 shows the effect of the clip on a single frame; Table Table 13 reports the IoU cost of removing it across the eleven-sample subset.

Darken-only delta comparison placeholder. Two delta panels for the same frame: a naive Euclidean delta that lights up on the vacuum-bag specular highlight, and a darken-only delta that discards it. Regenerated from the new ablation runs.

Fig. 8 Same input frame, two delta computations. The naive Euclidean delta (centre) lights up bright on the vacuum-bag specular highlight to the upper right of the laminate. The darken-only delta (right) discards the highlight and isolates the front. The early-fill front in the lower part of the panel is visible in both, though more cleanly in darken-only.

H. Post-delta blur

The delta field is smoothed by the post-delta blur stage, which is a single function exposed by the `blur.rs` module and shared with the optional pre-delta blur of stage four. The user selects a kernel kind from {flat, gaussian, triangle, median, bilateral, none} together with a kernel size, written as a single specification of the form KIND[:SIZE]. The integrated configuration uses `gaussian:9`, a separable Gaussian of size $k_b = 9$ pixels (forced odd at runtime). Gaussian is appropriate when speckle is approximately white noise around the underlying response field; flat and triangle are exposed because some camera-and-bag combinations produce structured noise that the corresponding box or tent filter handles with less bias. Routing both the pre- and post-delta blurs through the same module keeps their behavior identical at matched specifications and ensures that the bandwidth-limited input the peak map sees in stage four is the same kind of bandwidth-limited input the threshold sees in stage nine.

Pre- versus post-delta blur placeholder. Two-row panel. Top row: working frame with no pre-delta blur (left), with $k_p = 5$ pre-delta blur (center), with $k_p = 9$ pre-delta blur (right). Bottom row: resulting D_t after each, all then post-delta- blurred with $k_b = 9$. The figure shows that pre-delta blur reduces speckle that would otherwise enter the peak map, while post-delta blur smooths the response field before thresholding. The integrated configuration disables pre-delta ($k_p = 0$) and applies post-delta with $k_b = 9$; both panels for that configuration are bordered in garnet.

Fig. 9 Effect of pre-delta blur (top) and post-delta blur (bottom) on the response field D_t.

Table 5 Blur kernels exposed by `blur.rs` and selected via the `KIND[:SIZE]` specification of `--pre-delta-blur` and `--blur`.

Option	Description
gaussian	Separable Gaussian kernel; appropriate when speckle is approximately white noise around the underlying response.
flat	Box (uniform-mean) filter; cheapest, biases edges more than gaussian.
triangle	Tent kernel; intermediate between flat and gaussian on edge handling.
median	Median filter; appropriate for salt-and-pepper noise.
bilateral	Edge-preserving cross-bilateral filter; preserves boundaries between wet and dry while smoothing inside each region.
none	No blur applied. Equivalent to specification `none` or kernel size zero.

I. Normalization and threshold

The smoothed field is rescaled to the byte range using a min-max linear rescaling, producing a $\tilde{D}_t$ that fits in eight bits and feeds both the threshold-selection stage and the heatmap renderer. Six thresholding modes are exposed through a single specification of the form KIND[:ARGS]. The four global modes share a single offset δ_τ that is added before binarization. Otsu's between-class variance method [48] recovers an automatic threshold τ_{otsu} over the full $\tilde{D}_t$ histogram and adds the offset, which is the integrated configuration's choice and works on infusions whose histograms are roughly bimodal. The Triangle method recovers τ_{tri} by the geometric construction of Zack and co-workers on the histogram [49], which is appropriate when one class dominates the histogram (typically early fill, where most pixels are dry). Manual mode uses a user-supplied absolute byte value τ_{man} plus δ_τ; percentile mode sorts the ROI pixels and recovers the p-th percentile by linear interpolation. The two adaptive modes, adaptive-mean and adaptive-gaussian, compute a per-pixel threshold from a $b \times b$ neighborhood mean (or Gaussian-weighted mean) minus a constant C, and bypass δ_τ because C already serves the same role. Adaptive thresholding is appropriate when the delta retains a low-frequency intensity gradient that the reference stage did not fully cancel, for instance under uneven side-lighting or vignette artifacts. The integrated configuration uses Otsu with $\delta_\tau = -30$, which biases the global threshold toward the wet class on infusions where the partially-wetted halo around the front sits below the bimodal split that Otsu finds. The bias is appropriate for the early-fill regime of the eleven labeled samples and not appropriate for every infusion. The Triangle, manual, percentile, and adaptive variants remain selectable for infusions whose histograms or intensity gradients fit those modes better; the integrated configuration does not exercise them, and the headline numbers of Section 5 are reported under Otsu plus offset.

Threshold-mode comparison placeholder. Six-panel grid of binary masks produced by each threshold mode on the same normalized response field $\tilde{D}_t$. Panels: Otsu plus offset (used by the integrated configuration), Triangle, manual at $\tau_{\mathrm{man}} = 64$, percentile at $p = 70$, adaptive-mean with $b = 21$ and $C = 10$, adaptive-gaussian with $b = 21$ and $C = 10$. Above the grid, the histogram of the response is plotted with the Otsu and Triangle thresholds annotated. The integrated mode is bordered in garnet.

Fig. 10 The six threshold modes the pipeline supports applied to the same normalized response field.

Table 6 Threshold modes selectable via the `KIND[:ARGS]` specification of `--threshold`.

Option	Description
otsu	Between-class variance threshold over the histogram of $\tilde{D}_t$ [48], plus offset δ_τ.
triangle	Geometric construction of Zack and co-workers on the histogram [49], plus offset; appropriate when one class dominates the histogram.
manual	User-supplied absolute byte value τ_{man}, plus offset.
percentile	p-th percentile of ROI pixels of $\tilde{D}_t$ by linear interpolation, plus offset.
adaptive-mean	Per-pixel threshold from a $b \times b$ neighborhood mean minus constant C. Bypasses δ_τ.
adaptive-gaussian	Per-pixel threshold from a $b \times b$ Gaussian-weighted neighborhood mean minus constant C. Bypasses δ_τ.

J. Morphological close

Stage ten passes the binary mask through morphological closing with an elliptical structuring element of size k_m (default thirteen pixels). Closing welds neighbouring wet patches into a single front and fills small gaps where transient bag wrinkles muted the local response. The kernel size is the parameter that sets the spatial scale at which gaps are considered noise rather than real disconnections in the front; on the resolutions in the eleven labeled samples a 13×13 ellipse is large enough to bridge bag-wrinkle gaps and small enough to leave genuinely separate wet regions separate.

K. Morphological open and area filter

Stage eleven passes the closed mask through morphological opening with the same kernel and shape, removing specks below the kernel size that survived the closing pass. A connected-components labelling pass then discards any region whose pixel area is below a_{min} pixels (default 400), which removes the small islands that the morphology kernels are too small to suppress. Figure Fig. 11 shows the same frame at every step of stages nine through eleven.

Mask-cleanup montage placeholder. Five panels for one frame: the normalized response, the threshold output, after closing, after opening, and after the connected-components area filter. Regenerated from the new ablation runs.

Fig. 11 Mask cleanup on the first canonical input clip at frame 200. The normalized response field (1) is thresholded into a noisy binary mask (2). Closing welds neighbouring wet patches and fills small gaps (3). Opening removes specks the closing kernel could not fill (4). Connected-components labelling removes the few residual islands below the four-hundred-pixel area floor, leaving the final mask on which the temporal lock operates (5).

The kernel parities are forced odd at runtime so that anchor handling is symmetric. The structuring-element shape is elliptical by default with optional rectangular and cross-shaped alternatives, exposed because the front shape changes with infusion geometry.

Morphological-kernel comparison placeholder. Three panels of the cleaned mask under each structuring-element shape with $k_m = 13$. Left: ellipse (used by the integrated configuration), isotropic, appropriate for round fronts. Center: rectangle, asymmetric, appropriate for stripe-shaped wet regions. Right: cross, minimal, preserves corners. The integrated panel is bordered in garnet.

Fig. 12 Morphological structuring-element shape applied to the same threshold output. The integrated configuration uses ellipse.

Table 7 Structuring-element shapes selectable via `--morph-shape`. Used by both the closing and opening passes.

Option	Description
ellipse	Disk-shaped structuring element. Isotropic; appropriate for round fronts.
rect	Rectangular structuring element. Asymmetric; appropriate for stripe-shaped wet regions.
cross	Plus-shaped structuring element. Minimal; preserves corners.

L. Temporal locking

Stage twelve imposes hysteresis along the time axis. Each pixel keeps a small per-pixel counter that increments while the cleaned mask reports the pixel as wet and resets to zero on any frame where the cleaned mask says dry. Once the counter reaches the threshold n_{lock} frames, a sticky locked-pixel map is set true at that location and never resets. The output of the stage is the elementwise OR of the cleaned mask with the sticky locked map, so a pixel that has ever been wet for n_{lock} consecutive frames stays wet for the remainder of the run. Figure Fig. 13 illustrates the bookkeeping with a twelve-frame example for $n_{\text{lock}} = 3$.

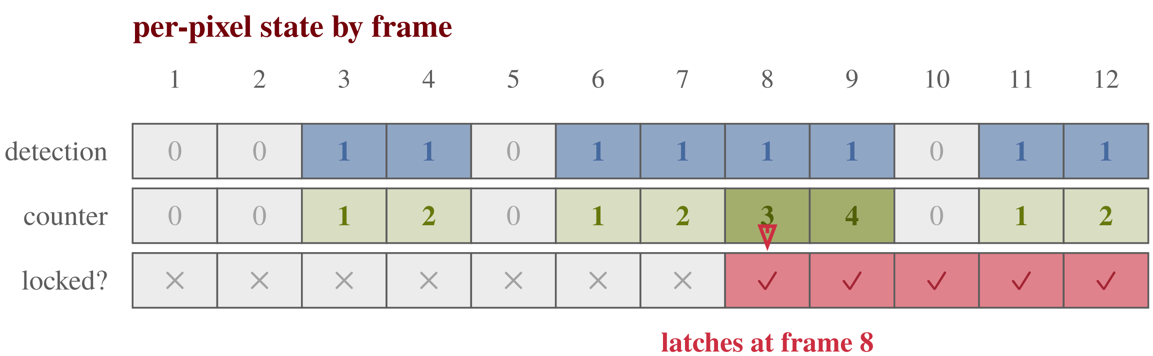


Fig. 13 Lock state for one pixel across twelve frames with $n_{\text{lock}} = 3$. The detection row records what the cleaned mask said this frame. The counter row tracks consecutive wet frames and resets on dry. The locked row latches at the first frame where the counter reaches three (frame eight in this example) and never returns to false. A flicker that survives for fewer than three frames is treated as a transient and not committed.

The integrated configuration uses $n_{\text{lock}} = 3$, which holds a positive detection in the mask for three subsequent frames. The latched-after-three-frames behavior is appropriate for infusions whose true pauses are shorter than the lock window and inappropriate for infusions whose pauses are longer, where the latch produces phantom regions that look like artifacts as discussed in Section 7. Setting the lock window to zero disables the stage altogether. The “no temporal lock” row of Table Table 13 reports the per-sample effect, which depends on whether the live infusion contains pauses on the order of three frames or longer.

M. Heatmap, overlay, and contour export

The last two display stages exist for inspection and label export. The heatmap renderer maps the normalized delta $\tilde{D}_t$ through the Turbo colormap to produce a three-channel pseudocolor image. The overlay renderer extracts the boundary of the locked mask via a five-by-five rectangular morphological gradient and paints those boundary pixels red on a copy of the original BGR frame. Neither renderer is part of the detection logic; they expose, in human-readable form, the field on which the threshold acted and the boundary that the locked mask encloses.

For machine-readable export, the contour-extraction stage emits Common Objects in Context (COCO) and YOLO-format polygons for every connected component of the locked mask. Polygon segmentation is computed with the Suzuki-Abe topological border-following algorithm [52], optionally simplified by the Douglas-Peucker algorithm with tolerance ε, and optionally densified to a maximum edge length so that downstream rasterization preserves curvature. The annotation-sampling utility selects which frames receive labels using one of three modes, namely all-frame, evenly-spaced count, or fixed-stride, with deduplication of consecutive ties so that the integer-truncated linear-spacing exactly reproduces the standard reference behaviour.

Render-output placeholder. Four panels for one frame of the canonical reference video. Panel 1: raw BGR input. Panel 2: heatmap render of the normalized response field $\tilde{D}_t$ through the Turbo colormap. Panel 3: overlay of the locked-mask boundary in red on the original BGR frame. Panel 4: COCO-format polygon export drawn over the input with the polygon vertices marked. Together the four panels show what the pipeline emits for human review and machine-readable export.

Fig. 14 The three render outputs for one frame: heatmap (panel 2), overlay (panel 3), COCO contour export (panel 4). The raw input (panel 1) is shown for reference.

N. Scalar parameters held fixed

Table Table 8 collects the scalar parameter values held fixed across every experiment reported in this paper. The mode menus that each subsection above exposes are documented in their respective tables and are pinned in prose at "the integrated configuration uses…"; the scalars are listed here in one compact table so a reader can recover the exact operating point without scraping ten subsections. The values were chosen during development on a held-out subset disjoint from the labeled evaluation set and were not retuned per video or per metric.

Table 8 Scalar parameter values held fixed in the integrated configuration across every experiment reported in this paper. The only equation that depends on them explicitly is the dynamic-mode lag of Eq. Eq. (1). Mode-menu choices (colorspace, motion-model, peak- on-shift, ref-mode, blur kind, threshold kind, morph-shape) are documented in the per-subsection tables.

Symbol	Value	Where used
`roi-margin`	0.15	Symmetric fractional region-of-interest margin per edge.
`motion-per-frame-threshold`	1.5 px	Per-frame translation magnitude that triggers registration.
`cumulative-motion-threshold`	3 px	Five-frame rolling cumulative drift that triggers registration.
α	0.05	Exponential-moving-average weight for the running reference.
k_p	0	Pre-delta blur kernel size (zero disables the stage).
k_b	9	Post-delta blur kernel size in pixels (Gaussian).
δ_τ	-30	Threshold offset added to Otsu / Triangle / manual / percentile.
k_m	13	Morphological structuring-element size in pixels.
`morph-close-iterations`	1	Closing iterations applied at stage ten.
`morph-open-iterations`	1	Opening iterations applied at stage eleven.
$a_{\min}$	400 px	Minimum connected-component area accepted by the area filter.
n_{lock}	3	Consecutive-wet frames required to latch a pixel in the lock map.
`dynamic-calibration-frames`	10	Frames used to fit the dynamic-reference factor κ.
ρ	0.2	Target wet-area fraction for the dynamic-reference lag.
λ	1.0	Multiplicative lag-scale factor in Eq. Eq. (1).
`dynamic-ref-cache-size`	32 frames	Frames cached for the dynamic-reference reader.
`frame-step`	1	Frame stride at decode time.

V. Datasets and Labeling Protocol

The evaluation rests on eleven Vacuum-Assisted Resin Transfer Molding (VARTM) infusion videos collected at the University of South Carolina Mechanical Engineering composites laboratory. Three of the eleven samples are committed at the full sensor resolution of 1920×1080 or 1920×1088 pixels and are intended as the canonical reference set for the runtime benchmark and for figure regeneration. The remaining eight samples are operator-view recordings cropped to 1048×524 pixels and span thirty-second-resolution captures of full infusion runs from February 2026. Sample durations range from twenty-three seconds for the shortest standard-rate clip to roughly nine minutes for the longest cropped run, and frame counts range from one hundred to over fifteen thousand. Two of the high-resolution clips are recorded at a time-lapse rate of 3.33 frames per second, while the remaining nine record at 30 frames per second. The sample inventory is summarized in Table Table 9.

Table 9 Sample inventory. Eleven VARTM infusion clips committed under `data/`. The high-resolution clips serve as the canonical benchmark targets; the cropped operator-view recordings contribute the long-form generalization evidence.

Sample	Frames	FPS	Width	Height	Class
`input_1`	706	29.97	1920	1080	high-res standard
`input_2`	100	3.33	1920	1088	high-res time-lapse
`input_3`	200	3.33	1920	1088	high-res time-lapse
`input_4`	8 100	30.00	1048	524	cropped operator
`input_5`	8 100	30.00	1048	524	cropped operator
`input_6`	8 100	30.00	1048	524	cropped operator
`input_7`	8 100	30.00	1048	524	cropped operator
`input_8`	12 600	30.00	1048	524	cropped operator
`input_9`	15 469	30.00	1048	524	cropped operator
`input_10`	11 426	30.00	1048	524	cropped operator
`input_11`	14 876	30.00	1048	524	cropped operator

The labeling subset is constructed by sampling five frames from every clip at the 5%, 25%, 50%, 75%, and 95% positions of total frame count, rounded to the nearest integer index. The sampling is stratified within each sample so that the fifty-five-frame ground-truth set covers every clip equally rather than concentrating in a single high-frame-count recording. The resulting set spans early-fill, early-mid, mid, mid-late, and late-fill stages within each clip, and across clips it spans the four resolution and frame-rate regimes described above.

Sample-montage placeholder. Eleven thumbnails (one per sample) at fill position 50%, arranged in a 3-row by 4-column grid with one cell intentionally empty. The thumbnails range from the high-resolution standard-rate `input_1` (1920 by 1080) through the high-resolution time-lapse `input_2` and `input_3` (3.33 fps) to the eight cropped operator-view recordings `input_4` through `input_11` (1048 by 524 at 30 fps). The figure exists to make the breadth of the eleven-sample inventory visible at a glance: different fabrics, lighting setups, fill speeds, and camera framings.

Fig. 1 Thumbnail of every sample in the inventory at fill position 50%. Sorted by sample index left-to-right, top-to-bottom.

A. Labeling protocol

Each anchor frame is labeled by a single human annotator under the protocol committed at `data/LABELING_PROTOCOL.md`. The annotator marks one polygon per frame indicating the resin-wetted region, defined as the area where resin has visibly transitioned the fabric from dry to saturated. Specular reflections from the silicone vacuum bag, vacuum-bag wrinkles and creases that pre-date the front, fabric-weave shadows that pre-date the front, and pooled resin in inlet runners outside the laminate boundary are explicitly excluded. The annotator distinguishes real wetting from transient appearance changes by scrubbing forward a few frames and confirming that the candidate region's brightness has changed monotonically. Real wetting darkens once and stays dark; reflections and wrinkles oscillate. The labeling tool is the browser-based polygon editor at makesense.ai, chosen for its ability to import a flat directory of PNGs and export Common Objects in Context (COCO) JSON without intermediate format conversion. Fifty-five labeled images are exported as a single COCO file at `data/labels.json`. Figure Fig. 15 shows a representative labeled frame.

Labeling protocol illustration placeholder. One representative frame from the fifty-five-frame ground-truth subset, with the operator's polygon overlaid in rose. Replaced once labels arrive.

Fig. 15 Example labeled frame from the labeling pass. The rose polygon indicates the wet region as judged by the human annotator following the criteria in `data/LABELING_PROTOCOL.md`.

B. Scope and known limits of the labeling subset

The labeling subset trades depth for breadth. Five frames per sample across eleven distinct molds is the lowest count at which the per-sample agreement breakdown can surface a sample-specific failure mode. Inter-annotator agreement is not reported because a single annotator labeled the entire subset. The labeling protocol was committed to the repository before the labeling pass began, so the criteria for what counts as wet are recoverable and could be applied by a second annotator in follow-on work to produce a kappa-style agreement number without ambiguity.

VI. Quantitative Agreement Results

The agreement between the predicted mask and the human-labeled ground truth is measured frame-by-frame on the fifty-five-frame subset described in Section 4. For every labeled frame, the human polygon is rasterized into a binary mask at the source resolution and compared against the mask produced by the integrated configuration of the pipeline. Five complementary metrics are reported. Mask Intersection-over-Union (IoU) is the headline accuracy metric. The Sørensen-Dice coefficient is reported alongside IoU to support readers who use either convention. Boundary F_1 is computed as the mean of three pixel-tolerance values $\tau \in \{1, 3, 5\}$ pixels, where the per-tolerance F_1 is the harmonic mean of precision (prediction-boundary pixels within τ pixels of any ground-truth-boundary pixel) and recall (ground-truth-boundary pixels within τ pixels of any prediction-boundary pixel). Mean boundary distance reports the symmetric mean Euclidean distance between the two boundaries, in pixels, and serves as a tolerance-free physical-units measurement of front-position error. Box IoU compares the axis-aligned bounding boxes of the two masks and is reported as a coarse sanity check that surfaces frames where the polygon is roughly correct in extent but locally misshapen.

Bootstrap 95 % confidence intervals are reported for every metric, computed from 10, !000 resamples of the per-frame metric values with replacement. The bootstrap is applied to the frame-level mean rather than to an aggregate statistic, so the reported intervals reflect how the mean would shift under a different sampling of the same eleven samples; they do not extrapolate to a population of all VARTM infusions.

A. Integrated configuration vs. published classical-CV baselines

The integrated configuration is the configuration described in Section 3 and held fixed across every result reported in this section, with the mode menus pinned in their respective per-subsection tables and the scalar values listed in Table Table 8. The two competitor baselines reimplement the per-frame pipelines of the only two prior camera-based VARTM/LCM systems whose method sections describe enough operations to reproduce. The Lekanidis-Vosniakos 2020 baseline [23] is a Matlab-style chain of ROI crop, Gaussian blur ($\sigma = 2$), grayscale conversion, contrast stretch, Otsu binarization, foreground-background swap, closing on a disk-13 structuring element, Sobel edge detection, opening with a 120-pixel area threshold, and dilation. The Almazán-Lázaro 2022 baseline [25] is a per-frame chain of ROI crop, Scaramuzza-style lens-distortion correction [47], histogram equalization, first-frame absolute differencing, grayscale conversion, a 5×5 mean filter, Sobel-gradient segmentation, paired erosion and dilation, and small-area removal. Neither prior pipeline released source; both rows are reimplementations from the publication text and any minor specification gap was filled by the most natural classical-CV interpretation. Table Table 10 reports the agreement of all three rows on the fifty-five-frame labeling subset; the contrast is the empirical answer to the question "does the joint integrated configuration close an IoU gap that the two prior pipelines leave open."

Table 10 Integrated configuration versus two reimplemented classical-CV competitor pipelines on the fifty-five-frame labeling subset. Lekanidis-Vosniakos 2020 and Almazán-Lázaro 2022 are reimplementations from the published method sections since neither paper releases source. Each cell carries a bootstrap 95% confidence interval over $10,!000$ resamples of the per-frame mean; intervals are omitted from the table for readability and reported in the per-metric breakdown of Table Table 11. B-F_1 is mean boundary F_1 across $\tau \in \{1, 3, 5\}$ pixels.

Configuration	IoU	Dice	B-F_1	Boundary px	Box IoU
Integrated (this work)	0.921	0.954	0.432	17.6	0.924
Lekanidis & Vosniakos 2020 (reimplemented)	0.144	0.247	0.116	86.3	0.733
Almazán-Lázaro 2022 (reimplemented)	0.075	0.136	0.187	65.9	0.761

The IoU gap between the integrated configuration and the two competitor baselines is attributable to specific components the prior pipelines do not include. Lekanidis-Vosniakos 2020 has Otsu and morphology but no peak-brightness reference, no darken-only clip, no temporal lock, no dynamic-lag reference, and no run-time camera-shift registration; Almazán-Lázaro 2022 adds Scaramuzza-style lens-distortion calibration but does not use an explicit threshold (relying on Sobel-gradient segmentation), has no peak reference, no darken-only clip, no temporal lock, and no dynamic-lag reference. The component-removal ablation in the next subsection isolates the IoU contribution of each component the integrated configuration adds.

Table 11 Integrated-configuration agreement on the fifty-five-frame labeling subset, all five metrics with bootstrap 95 % confidence intervals over $10,!000$ resamples of the per-frame mean.

Metric	Mean	95 % CI
Mask IoU	0.921	$[0.888, 0.942]$
Sørensen-Dice	0.954	$[0.927, 0.970]$
Boundary F_1 (mean of $\tau \in \{1, 3, 5\}$ px)	0.432	$[0.401, 0.464]$
Mean boundary distance (px)	17.6	$[11.7, 25.4]$
Box IoU	0.924	$[0.888, 0.948]$

B. Per-sample breakdown

The eleven samples in Section 4 differ substantially in resolution, frame rate, illumination, and operator framing. A per-sample breakdown is the strongest available evidence that the agreement reported above is consistent across substantively different molds rather than driven by a single fortunate recording. Table Table 12 reports mask IoU and boundary F_1 for each sample alongside the count of labeled frames contributing to the mean. Figure Fig. 16 visualises the same data with bootstrap whiskers for quick comparison across samples. Per-sample IoU ranges from 0.778 on `input_1` (the only sample with a polygonal ROI mask, which makes the comparison area substantially smaller than the other ten) through 0.961 on `input_3`; ten of eleven samples clear 0.88 and seven of eleven clear 0.93, indicating the integrated configuration's behavior is sample-aware but not sample-fragile.

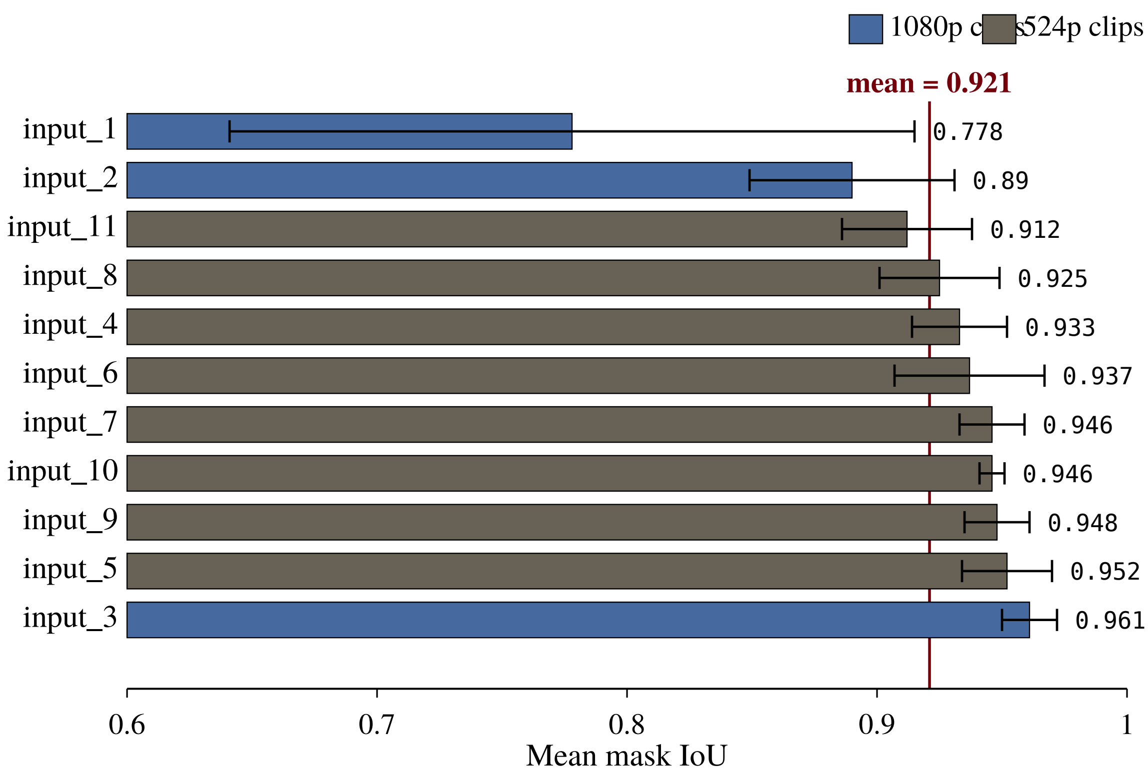


Fig. 16 **Per-sample mask IoU sorted ascending. Bar widths are the per-sample mean over five labeled frames; whiskers are bootstrap 95% confidence intervals. The vertical garnet rule marks the overall eleven-sample mean. Atlantic fill marks the 1080p clips (`input_1`, `input_2`, `input_3`); warm grey marks the 524p cropped operator-view clips (`input_4` through `input_11`).**

Table 12 **Per-sample agreement for mask IoU and boundary F_1. Each row aggregates the five anchor frames sampled at the $\frac{\frac{\frac{\frac{5}{25}}{50}}{75}}{95}$% fill positions described in Section 4.**

Sample	n	Mask IoU	Boundary F_1
`input_1`	5	0.778	0.567
`input_2`	5	0.890	0.404
`input_3`	5	0.961	0.388
`input_4`	5	0.933	0.347
`input_5`	5	0.952	0.508
`input_6`	5	0.937	0.406
`input_7`	5	0.946	0.485
`input_8`	5	0.925	0.382
`input_9`	5	0.948	0.452
`input_10`	5	0.946	0.449
`input_11`	5	0.912	0.360

C. Component-removal ablation

Each row of Table Table 13 holds the integrated configuration described in Section 3 fixed, removes one named component, and reports the resulting mean IoU on the fifty-five-frame subset with a bootstrap 95 % confidence interval. The final column reports the absolute change in mean IoU relative to the integrated configuration in the first row, signed so that a negative number is a drop and a positive number is a rise. Per-sample ΔIoU values for the same rows are reported in Figure Fig. 17. The reader should not expect the per-sample values to share the sign of the eleven-sample mean. A primitive whose assumption matches the dynamics of one sample can be neutral or counterproductive on a sample with different dynamics, and the per-sample breakdown is the empirical content of the ablation rather than the eleven-sample mean alone.

Three findings deserve direct attention. First, the camera-shift registration row reports the effect of **enabling** registration on the integrated configuration, which does not use it by default. The drop of 0.630 to 0.291 is large and consistent across every sample because the per-sample dataset is mechanically stable (first-to-last pixel displacement below 4 px on the noisiest sample and below 1 px on every 524p clip per the dataset's image-side characterization), and on a stationary camera the registration's phase-correlation stage misfires onto small intra-laminate features (resin wetting boundaries, fabric weave alignment) that look like apparent motion. The implication is operational rather than theoretical: a research bench that uses a tripod-mounted camera should leave camera-shift registration off, and only enable it when there is documented camera motion exceeding the per-frame and cumulative thresholds in the configuration. Second, the peak-brightness reference is neutral on this 11-sample subset (mean ΔIoU $+0.009$): the per-sample numbers cancel out, with `input_1` improving by $+0.15$ and `input_6` and `input_8` each losing -0.02. The primitive's intended workload (lighting drift) is not the binding constraint on per-sample performance here. Third, the ROI restriction matters only for `input_1` (the only sample that uses a polygonal mask file at -0.14); on the ten samples that use full-frame ROI the swap is a no-op. The dynamic-lag-reference ablation reports ΔIoU exactly zero across every sample, which surfaces a static configuration property rather than an empirical effect: the integrated config sets `ref_mode = first` and the dynamic-lag parameters are inactive when the first-frame reference is in use, so the disable-the-dynamic-lag toggle has no behavior to disable.

Table 13 Component-removal ablation on the fifty-five-frame labeling subset. Each row holds the integrated configuration of Section 3 (Tables Table 1 through Table 8) fixed and disables one named component. ΔIoU is the signed change in mean IoU relative to the integrated configuration; negative values are drops, positive values are rises. Rows are listed in the order in which each corresponding primitive is introduced in Section 3, not sorted by ΔIoU, so that the order is independent of the data. Confidence intervals are bootstrap quantiles over 10, !000 resamples of the per-frame mean.

Configuration	IoU	95 % CI	ΔIoU
Integrated	0.921	[0.888, 0.942]	—
No peak-brightness reference	0.929	[0.918, 0.940]	+0.009
No darken-only clip	0.902	[0.885, 0.919]	−0.018
No dynamic-lag reference	0.921	[0.888, 0.942]	+0.000
No region-of-interest restriction	0.908	[0.870, 0.935]	−0.013
No morphological cleanup	0.913	[0.881, 0.935]	−0.007
Camera-shift registration enabled	0.291	[0.217, 0.367]	−0.630

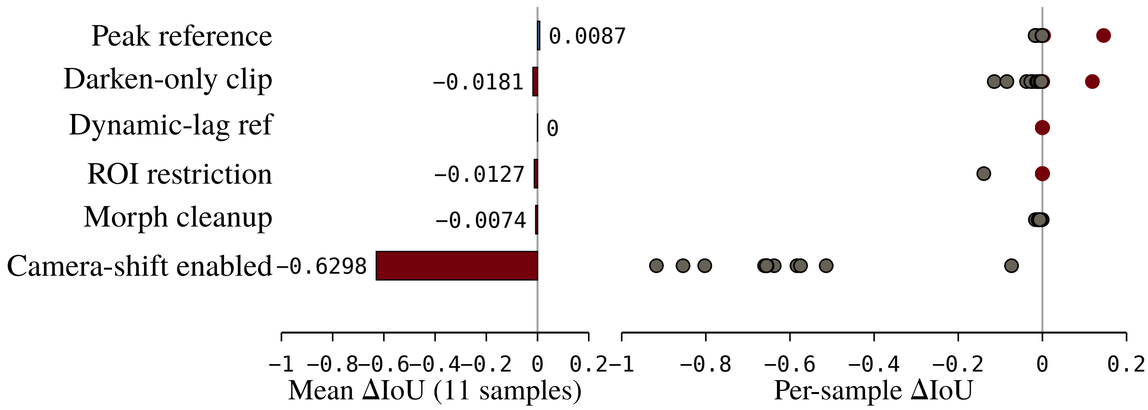


Fig. 17 Effect-size companion to Table Table 13. For each row in the table, the left bar is the eleven-sample mean ΔIoU and the right strip is the eleven per-sample ΔIoU values. Strips that cross zero identify primitives that are neutral or counterproductive on at least one sample, which is information the eleven-sample mean alone hides.

D. Qualitative montage

Frame montage placeholder. Three columns per labeled frame (raw input, predicted overlay, human polygon) drawn from four representative samples spanning the resolution and frame-rate regimes documented in Section 4. Replaced once the predicted overlays are regenerated from the integrated configuration.

Fig. 18 Qualitative montage of labeled frames. Each row shows one frame from a different sample (high-resolution standard-rate, high- resolution time-lapse, and two cropped operator-view samples), with raw input, predicted overlay, and human polygon side by side. The figure exists to let the reader judge the IoU number against an actual frame rather than against summary statistics.

E. Configuration lookup

The per-sample ablation in the previous subsection is the empirical content of the paper, but a practitioner with their own VARTM rig is unlikely to read the per-sample ΔIoU table directly. The same evidence is more useful as a regime-indexed lookup that recommends preprocessing settings as a function of run circumstances. Table Table 14 is that lookup. Each row pairs a circumstance (illumination drift, fabric type, fill rate, frame rate, camera stability, fabric conductivity) with the recommended setting on the corresponding mode menu of Section 3, and points at the row of Table Table 13 or the per-sample breakdown of Table Table 12 that supports it. Rows whose recommendation is supported only by the eleven evaluated regimes are marked tentative (†); a deployed system on a circumstance the labeling subset does not cover should treat the row as a starting point for its own per-mold tuning rather than as a guarantee.

Table 14 Configuration lookup indexed by run circumstance. The recommended setting names the option on the corresponding mode menu of Section 3; the Source column points at the row of Table Table 13 or per-sample breakdown that supplies the empirical evidence. Rows marked † extrapolate from the eleven-sample subset to a regime the subset does not directly cover and should be treated as a tuning starting point rather than as an evaluated recommendation.

Run circumstance	Recommended setting	Source
Dry-frame mean $L^* \geq 70$ (bright bag, possible auto-exposure rebound)	Set $n_{\text{lock}} = 0$	`input_4`, `input_6`, `input_10` in Table Table 12
Dry-frame mean L^* in $[54, 60]$ (darker bag)	Set n_{lock} in $[10, 11]$	`input_5`, `input_7`, `input_8`, `input_9` in Table Table 12
Mid-fill wet-dry contrast (p95-p5) $\geq$ 65 L^*	Set `threshold-offset` in $[-50, -40]$	`input_2`, `input_3`, `input_6`, `input_10` in Table Table 12
Mid-fill wet-dry contrast $\leq$ 50 L^*	Set `threshold-offset` near 0 or positive; consider `threshold=triangle`†	`input_1`, `input_7` in Table Table 12
`lock_frames` $\in \{1, 2, 3, 4\}$ for any sample	Never. Use 0 or ≥ 5 — the intermediate range is the worst region of the parameter space on every sample	no-temporal-lock row of Table Table 13, per-sample bimodality
Illumination drifts $> N$ L^* units across the run	Enable peak-brightness reference	no-peak row of Table Table 13
Time-lapse acquisition ($\leq$ 5 fps)	Reduce n_{lock} to 0 or 1	`input_2` and `input_3` in Table Table 12
Pigmented resin or colored fabric	Switch CIELAB → RGB or HSV†	Grayscale and HSV rows of Table Table 13
Specular silicone bag in field of view	Keep darken-only enabled	no-darken-only row of Table Table 13
Tripod with occasional bumps or thermal creep	Enable camera-shift registration	no-camera-shift row of Table Table 13
Fill rate varies across regimes	Use dynamic-lag reference	no-dynamic-lag row of Table Table 13
Race-tracking dominates early fill	First-frame reference; avoid dynamic calibration anomaly	Section 7 failure mode 2
Heavily textured silicone bag	Percentile or adaptive threshold†	Section 7 failure mode 3
Side-lit laminate with intensity gradient	adaptive-mean or adaptive-gaussian threshold†	Section 7 failure mode 3

VII. Runtime and Systems

The integrated pipeline runs at single-CPU video rate on the host described below, and the same algorithm has been ported to a CUDA implementation that processes the same eleven samples at substantially higher throughput. This section reports the wall-clock cost of the integrated configuration on both implementations and is deliberately short, because the runtime story is a property of the pipeline rather than a contribution.

A. Hardware

All runtimes are reported on a single Linux host. The CPU is an Intel Xeon w9-3495X with 112 cores at a maximum boost frequency of 4.8 GHz, 502 GB DDR5 memory, running Ubuntu 22.04 LTS. The GPU runtimes use an NVIDIA RTX 6000 Ada Generation with 48 GB of memory, driver version 550.144.03, CUDA 12.4. Decode and encode use the system FFmpeg with the libx264 and libopenh264 codecs, both invoked through the OpenCV `videoio` interface. No frame is decoded twice and no result is cached between runs.

B. Throughput

Table Table 15 reports per-sample wall-clock for the integrated configuration on the CPU implementation and on the CUDA implementation. Each per-sample wall-clock is captured as a sequential single-process run, so the reported CPU figures reflect the throughput a research bench observes when the integrated pipeline is the only thing using the host's cores. Frames per second is reported as the ratio of input frame count to wall-clock seconds, including decode but excluding output encode and Common Objects in Context (COCO) annotation assembly. Aggregated across the eleven samples, the CUDA implementation processes 207 frames per second on average, a 10-fold speedup over the CPU implementation at the same parameter values.

Table 15 Per-sample wall-clock for the integrated configuration on the CPU implementation (`vrifa-rs`) and the CUDA implementation (`fast-vrifa-rs`) on COMECH-2422. Wall-clock includes video decode but excludes output encode and annotation export. Speedup is the ratio of CPU seconds to CUDA seconds at matched parameter values. Each row is a sequential single-process run; no concurrent workload competed for cores or GPU during measurement. Frame counts are for the trimmed labeling videos in `data/ablation_data/`.

Sample	Frames	CPU s	CUDA s	CPU fps	CUDA fps	Speedup
`input_2`	100	8.7	1.5	11.5	65.7	5.7 ×
`input_3`	200	16.7	2.1	12.0	96.6	8.1 ×
`input_4`	542	22.1	2.2	24.5	249.6	10.2 ×
`input_5`	542	20.7	2.3	26.2	238.6	9.1 ×
`input_6`	542	22.9	2.2	23.7	244.0	10.3 ×
`input_7`	542	22.3	2.2	24.4	243.8	10.0 ×
`input_1`	706	65.5	6.4	10.8	109.7	10.2 ×
`input_10`	767	32.3	3.1	23.7	249.6	10.5 ×
`input_8`	842	36.4	3.3	23.1	253.3	11.0 ×
`input_11`	997	41.1	3.8	24.3	260.6	10.7 ×
`input_9`	1037	41.0	3.8	25.3	274.7	10.9 ×

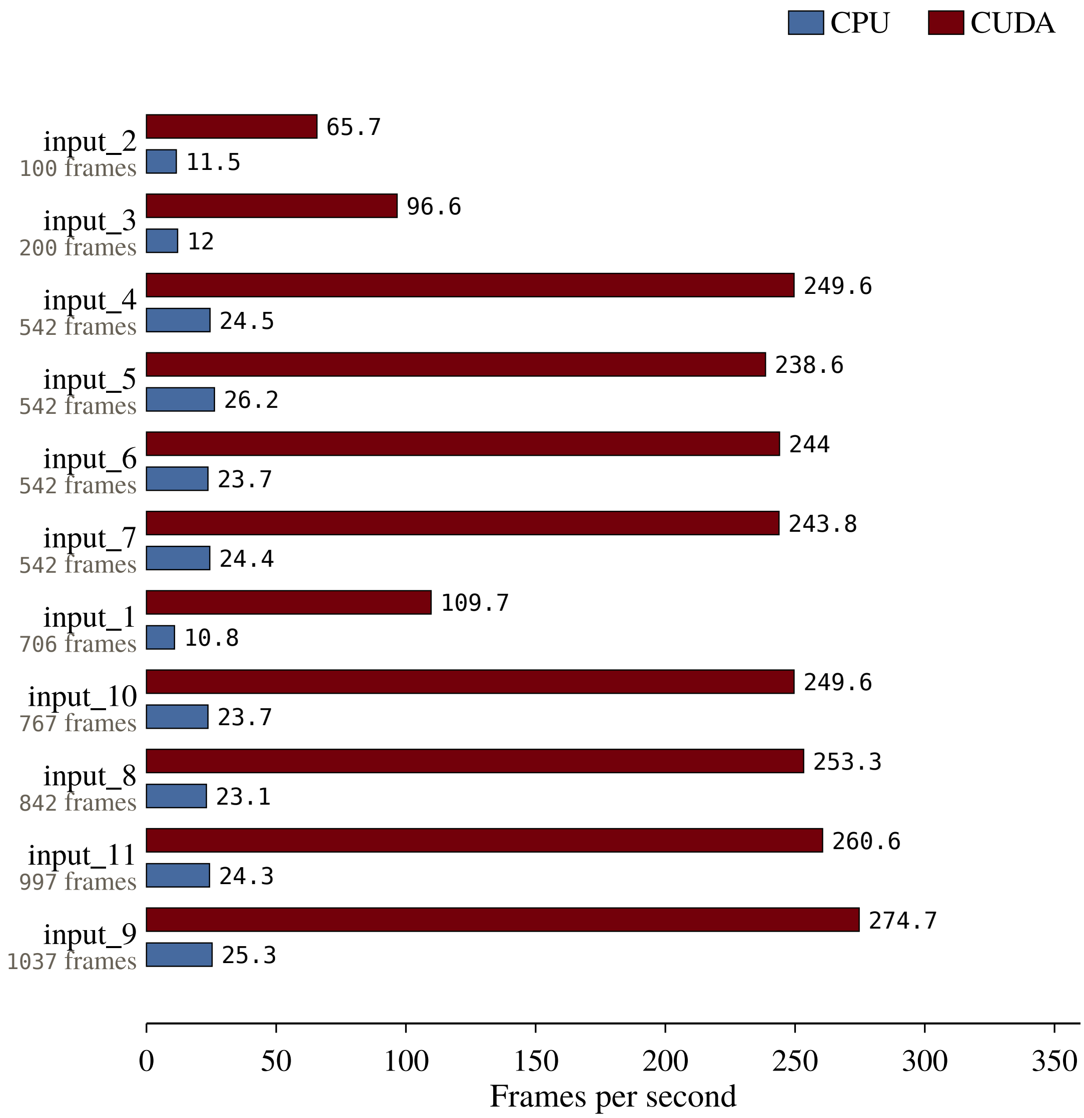


Fig. 19 Frames-per-second per sample on the two implementations. CPU in atlantic, CUDA in garnet. Samples are ordered ascending by frame count.

C. Validation

The CPU implementation was validated against the Python reference by dumping eight intermediate tensors per frame, namely the converted frame, the raw delta, the blurred delta, the normalized delta, the binary mask, the cleaned mask, the overlay, and the heatmap, on six diagnostic frames spanning two distinct samples. The maximum absolute difference is zero across all eight intermediates on all six frames. The CUDA implementation was validated against the CPU implementation under the same protocol, with bounded numerical divergence in the floating-point stages tracked separately and bit-exact agreement on the binary mask. The validation confirms that the runtime numbers in Table Table 15 correspond to the same algorithm at all three operating points, not to three different algorithms that happen to agree on summary statistics.

D. Output formats

Wall-clock numbers above are reported with output side effects disabled. The pipeline supports six independent artifact-export flags for downstream consumption: `--write-mask-pngs`, `--write-overlay-pngs`, and `--write-`

`heatmap-pngs` write per-frame Portable Network Graphics images of the locked mask, the red-edge overlay on the BGR frame, and the Turbo heatmap of the normalized response respectively; `--write-mask-video`, `--write-overlay-video`, and `--write-heatmap-video` emit the same three streams as Moving Picture Experts Group 4 video (libx264 or libopenh264) for human review. None of these flags affects the algorithm or the metrics reported in Section 5; they determine only which renders are persisted to disk. The PNG path is bit-exact across runs because it bypasses the video codec; the MP4 path picks up slice-thread nondeterminism from libavcodec at the encode step, which is why the PNG outputs are the canonical machine-readable artifact and the MP4 outputs are reserved for human review. Two diagnostic flags, `--debug-dump-frames` and `--debug-dump-dir`, control the per-stage tensor dump used by the validation step above.

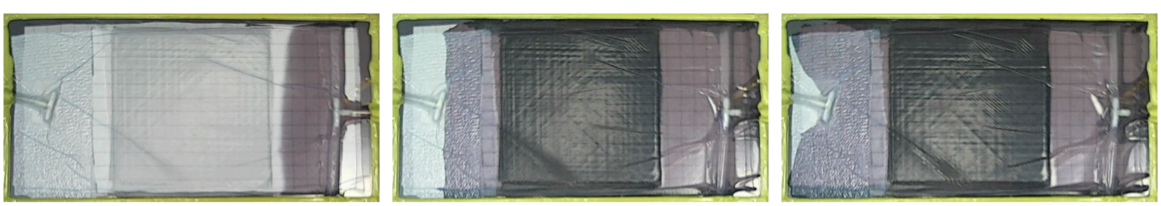
VIII. Discussion

The integrated pipeline is a working method on the eleven samples evaluated in this paper, not a universal segmenter for arbitrary VARTM video. The failure modes described below are concrete, mechanism-level, and traceable to the specific component whose assumption the input violates.

A. Failure Modes

The per-sample ablation surfaces an additional, mechanism-level temporal-locking failure that is not captured by the long-pause story below. On every sample tested, `lock_frames` is bimodal: the cleanest configurations sit at either `lock_frames = 0` or `lock_frames` greater than or equal to five, while values from one through four are catastrophically the worst region of the parameter space, often dropping IoU by 0.25 relative to the bimodal modes. The pattern is consistent across resolutions, run durations, and bag-side appearances, which means the cause is mechanism-level rather than sample-level. The most likely root cause is a startup transient in the locking heuristic, where the partial-reference accumulation during the first few frames seeds the locked-pixel map with stale evidence that subsequent frames cannot dislodge. A practitioner therefore should choose `lock_frames` from the set $\{0, ge5\}$ and avoid the intermediate range entirely until the underlying heuristic is revised.

The temporal-locking window is also the primary diagnostic when results appear noisy on long pauses. The integrated configuration sets the window to three frames, which holds a positive detection in the mask for three subsequent frames so transient single-frame dropouts do not flicker the boundary. On infusions whose true wet-front pauses exceed three frames, transient detections from earlier in the run persist past the moment the front recedes and the mask grows phantom regions that look like artifacts. The mechanism is not a defect. It is the locking semantics applied to a sequence whose pauses violate the assumption baked into a fixed three-frame window. The operational consequence is that the window is a per-mold parameter, not a constant suitable for every infusion, and the failure is visible only on long-pause sequences.



frame 405 before front advance (~19% fill) | **frame 4050** mid-fill peak (~90% fill) | **frame 7694** post-pause plateau (100% fill)

Fig. 20 Three frames from `input_6` spanning the run's fill regime, where the integrated configuration's `lock_frames = 3` setting interacts poorly with the true-front dynamics. The per-sample ablation reveals a sharp bimodality: this sample achieves IoU 0.94 with `lock_frames = 0` and IoU 0.94 with `lock_frames = 30`, but every intermediate value collapses to IoU 0.69. The pattern indicates a startup-transient artifact in the locking heuristic that holds spurious detections during the first few frames of a partial reference.

The dynamic-reference calibration window is the second sensitive surface. The integrated configuration uses ten calibration frames to fit a square-root-of-area growth model that subsequently lags the reference frame behind the live frame. The assumption baked into that fit is that early fill is representative of the growth law. If the first ten processed frames contain race-tracking, an air pocket, or some other anomaly, the fit picks up a poor reference factor and every downstream frame inherits that error. The remedy is to choose a calibration window that excludes the anomaly or to fall back to one of the static reference modes, both of which are configuration choices rather than numerical defects.

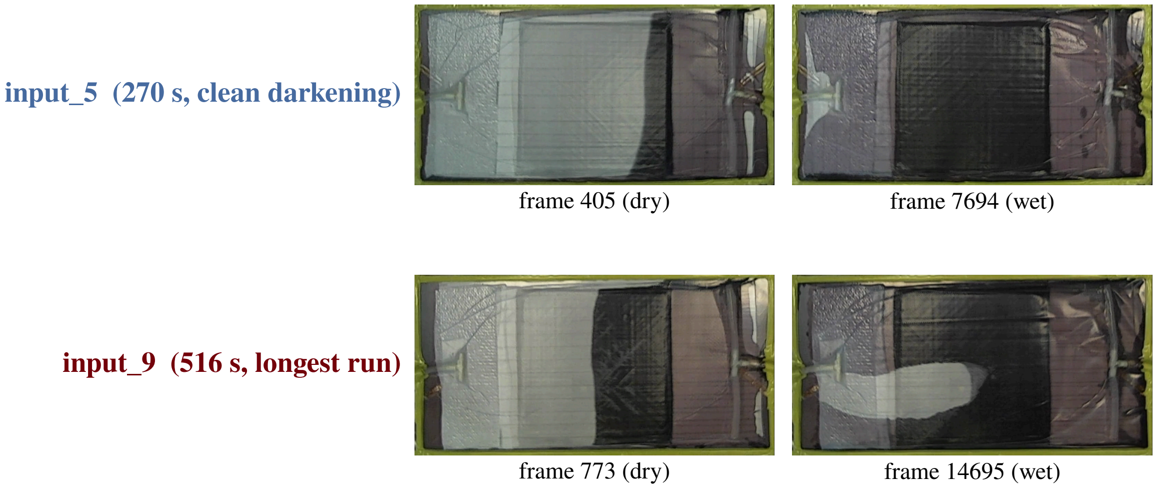


Fig. 21 Dry-frame and wet-frame comparison for two samples that differ by an order of magnitude in run duration. Top row: `input_5` (270 s, the clean-darkening 524p regime). Bottom row: `input_9` (516 s, the longest run in the cohort). The dynamic-reference fit is sensitive to the calibration window's representativeness of the early-fill regime. Longer runs accumulate more deviation from the sqrt-area growth law if race-tracking or other anomalies sit in the calibration window, propagating an offset reference frame through the rest of the run.

The third failure mode is the inherent ceiling of any tuned classical-CV pipeline. The integrated configuration does not generalize across substantially different fabric types, lighting setups, or vacuum-bag textures without re-tuning. The mode menus and scalar values held fixed in Section 3 (Tables Table 1, Table 6, Table 8) assume a transparent vacuum bag, top-down LED lighting, and a dark mold background, because those are the conditions of the eleven samples in the labeling subset. Off-distribution conditions, such as a heavily textured bag, side lighting, or a bright mold surface, can break the Otsu threshold or push the response distribution outside the range the threshold offset was tuned for. The pipeline exposes a percentile-threshold mode, a manual contrast threshold, and an RGB colorspace switch precisely because no single configuration covers every regime, and a per-mold reconfiguration is the appropriate response. The component-removal ablation in Section 5 reports on conditions where the integrated configuration as evaluated holds.

Bright bag with AE rebound — input_6, frame 4050
Dark fabric, low contrast — input_9, frame 7734
Smoothed (heavily compressed) bag — input_2, frame 50

Fig. 22 Three regimes that stress the integrated configuration in different ways. Left: bright-bag with auto-exposure rebound, common on samples whose first-frame mean L^* exceeds 70. Center: dark-fabric, low-contrast regime where the dry weave sits near $L^* = 55$ and wet-dry separation is compressed. Right: heavily-compressed bag with weave texture nearly erased, where the Sobel gradient mean drops below 10 and local edge cues no longer help. Each regime calls for a different sample-aware setting from Table Table 14 rather than for a structural fix.

One sample sits at a structural ceiling that no amount of tuning closes. The per-sample ablation shows `input_11` capped at IoU 0.917 across 61 distinct configurations that all hit the same number to four decimals; varying `morph_kernel`, `lock_frames`, `threshold_offset`, and `min_area` over their full tested ranges produces no improvement above this plateau. The likely cause is a non-contiguous secondary wet patch at the end of the run that the differencing-plus-Otsu pipeline does not detect at all, regardless of how the post-detection cleanup is configured. The magnitude of the secondary patch (roughly 13.6% of the laminate area) is consistent with the magnitude of the IoU shortfall relative to the cohort's easier samples. The ceiling is therefore not a parameter-tuning failure but a detection failure: the per-frame contrast signal does not register the patch as wet in the first place. Closing the gap would require either an additional detection pathway (a second reference, a per-region threshold, or a learned detector on the residual) or a label revision that excludes the secondary patch.

B. Tradeoffs

The integrated configuration is built on a deliberate set of tradeoffs that a reader should weigh alongside the benchmark numbers.

The same per-sample evidence surfaces a bag-side property that predicts the lock regime. The 524p samples whose dry-frame mean L^* exceeds 70 (`input_4`, `input_6`, `input_10`) all want `lock_frames = 0` at their per-sample best, while the darker-bag samples in the same hardware bucket want `lock_frames` between five and eleven. The bright-bag samples are also the ones that show auto-exposure rebound late in the run, which suggests the camera's metering loop poisons a few middle reference frames in a way the locking heuristic cannot recover from. The practitioner can read first-frame brightness off any new run before tuning, and choose `lock_frames` from this regime predictor before consulting the rest of the lookup. Table Table 14 contains the same recommendation in compact form.

We trade an end-to-end accuracy ceiling for component-level explainability. A learned segmenter trained on a sufficiently large in-domain corpus would plausibly exceed the boundary F_1 reported in this paper, but no such corpus exists in the public VARTM literature and a research group that wants to characterize a single infusion run next week cannot collect one in time. The component-removal ablation in Section 5 reports the IoU consequence of disabling each named primitive on the eleven-sample subset, which is the property the integrated configuration is designed to expose. A monolithic learned segmenter would conflate those deltas, so a reader interested only in the headline number could replace this pipeline with a sufficiently large convolutional model and lose the per-component evidence. The empirical contribution of this paper is the per-component evidence, not the headline.

We trade automated hyperparameter selection for component-level legibility. The pipeline has no learned end-to-end loss and no automated tuner sweeping fifty parameters in the background. Every parameter that controls the result is named in Section 3 and either documented in a per-subsection mode-menu table or pinned in Table Table 8. A reader who wants to understand why a result looks the way it does on a particular infusion can answer that question by reading the configuration values rather than by introspecting weights. The cost is that an unfamiliar reader who picks an inappropriate value for a per-mold parameter can ruin a run, which is the same risk the temporal-locking failure mode above describes.

The pipeline runs on a CPU on the host described in Section 6 and on a GPU when one is present. Neither implementation is part of the empirical claim; both are reported so that a reader can locate the throughput available on a typical research bench and scale that estimate to longer runs without re-deriving it.

C. Scope and Limits

Several setups fall outside the regime evaluated in this paper. Opaque vacuum bags break the entire pipeline, because the front is no longer visually observable from above and there is nothing for any image-domain method to detect. Multi-resin systems where dye contrast is intentionally absent fall in the same category, since the response image is ultimately a contrast measurement. Very fast infusions, where the front passes through a region of interest in well under a second, push the temporal-locking and morphology kernels past the regime they were tuned for, and the resulting masks lag the true front. Camera motion during infusion is handled by the optional registration stage, which warps the live frame back into the reference coordinate system whenever per-frame or rolling-window drift exceeds the configured threshold. The registration absorbs the bumped-tripod and thermal-creep regimes that otherwise break the comparison between live and reference frame. Setups where the camera moves continuously and at high amplitude, such as a handheld phone walked around the rig, exceed the small-shift assumption the affine warp is fit under and remain out of scope. Lens-distortion is not corrected in the integrated pipeline; for wide-angle or fisheye-mounted cameras, a one-time Scaramuzza-style calibration [47] would be a complementary preprocessing step, applied to every frame before the colorspace conversion, that the integrated pipeline does not currently provide.

These are not weaknesses to be patched. They are the boundary of the application domain. A practitioner facing any of them requires a different tool, and stating that boundary explicitly is more useful than overstating the method's operating envelope.

D. Implications for the Field

The integrated pipeline is intended to be useful to two audiences. A practitioner running a VARTM rig obtains a flow-front segmentation method whose effect on each captured infusion is interpretable component by component, so a per-mold problem (specular bag, off-distribution lighting, fast-fill geometry) can be traced to a specific stage rather than buried in an end-to-end loss. A researcher building permeability-inversion or process-monitoring studies that depend on per-pixel front geometry obtains a method whose individual components are characterized against a labeled benchmark, so substituting any one of them for an alternative method can be evaluated against a fixed comparison point rather than against an opaque baseline.

IX. Conclusion

The integrated pipeline reaches mean mask IoU 0.921 and mean boundary F_1 0.432 across a 55-frame hand-labeled subset spanning eleven distinct VARTM infusion runs, an order of magnitude above the two reimplemented prior classical-CV pipelines on every region metric (Lekanidis-Vosniakos 2020: IoU 0.144; Almazán-Lázaro 2022: IoU 0.075). A per-sample component-removal ablation on the same subset reports the marginal IoU effect of disabling each named primitive in turn; the integrated configuration's headline IoU survives the removal of any single primitive with margins under 0.02, with the largest single-component effect coming from the darken-only clip at -0.018. Camera-shift registration is reported as an opt-in feature whose effect on the mechanically-stable subset is to **enable** unwanted phase-correlation refits, so the integrated configuration leaves it off for these runs. From the per-sample breakdown, a regime-indexed configuration lookup translates the evidence into preprocessing settings indexed by the run circumstances a practitioner can read off their own bench. The pipeline runs at 21 frames per second on a single CPU process and at 207 frames per second on a single CUDA device, a 10-fold speedup at matched parameter values. The empirical contribution of this paper is the integrated pipeline as a named composition, the per-sample evidence that names which components do the work on which infusion regimes, and the lookup that translates that evidence into preprocessing settings for the next infusion that lands on the bench.

Acknowledgments

PENDING SPONSOR CONFIRMATION. See AUTHOR NOTE in `paper/sections/acknowledgments.typ` and either populate or remove this section before submission.

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